SQL BASICS

Q9) What is the Difference Between a Primary Key and a Foreign Key?

1️⃣ Primary Key

Definition:
A **Primary Key (PK)** is a column (or combination of columns) in a table that **uniquely identifies each row** in that table.

Key Points:

- Must be **unique** — no two rows can have the same primary key value.
- Cannot be **NULL** — every row must have a value.
- Usually, one table has **only one primary key**.
- Often used to link other tables via foreign keys.

2️⃣ Foreign Key

Definition:
A **Foreign Key (FK)** is a column (or set of columns) in one table that **refers to the Primary Key of another table**. It establishes a **relationship between two tables**.

Key Points:

- Can have **duplicate values** — multiple rows in one table can reference the same row in another table.
- Can be **NULL** if optional.
- Enforces **referential integrity** — ensures data in the child table matches a valid row in the parent table.

Feature	Primary Key	Foreign Key
Purpose	Uniquely identifies a row	References a row in another table
Uniqueness	Must be unique	Can have duplicates
Nulls	Cannot be NULL	Can be NULL (optional)
Table	One per table	Can have multiple FKs per table
Integrity	Row-level uniqueness	Referential integrity across tables

✅ **Summary:**

- **Primary Key** → "Who am I?" (unique identity for a row)
- **Foreign Key** → "Who am I connected to?" (link to another table)

Q10) What Are Constraints in SQL and Why Are They Used?

1️⃣ What Are Constraints?

Constraints are **rules applied to columns or tables** in a database to **enforce data integrity and ensure valid data**.

They restrict the type of data that can be inserted, updated, or deleted in a table.

2️⃣ Common Types of Constraints

Constraint	Purpose
PRIMARY KEY	Uniquely identifies each row in a table; cannot be NULL
FOREIGN KEY	Ensures a column's value matches a value in another table (maintains relationships)
UNIQUE	Ensures all values in a column are unique (no duplicates)
NOT NULL	Prevents NULL values in a column
CHECK	Ensures values in a column satisfy a specific condition (e.g., salary > 0)
DEFAULT	Assigns a default value if no value is provided during insert

3️⃣ Why Are Constraints Used?

Constraints are used to:

1. **Maintain Data Integrity** – ensure data is accurate and consistent.
2. **Prevent Invalid Data** – for example, you cannot insert a sale with a negative amount if you have a CHECK constraint.
3. **Enforce Relationships** – foreign keys maintain links between related tables.
4. **Simplify Queries** – knowing data rules helps avoid errors when querying or joining tables.
5. **Reduce Application-Level Validation** – constraints shift some data validation to the database itself.

Example: Employees Table with Constraints

```
CREATE TABLE employees (
  employee_id INT PRIMARY KEY,
  first_name VARCHAR(50) NOT NULL,
  last_name VARCHAR(50) NOT NULL,
  department VARCHAR(50),
  salary INT CHECK(salary > 0));
```

Explanation:

- PRIMARY KEY → unique employee ID
- NOT NULL → first and last names cannot be empty
- CHECK(salary > 0) → prevents negative salary values